

3.1 An Introduction to AS Level Organic Chemistry

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.1 An Introduction to AS Level Organic Chemistry
Difficulty	Easy

Time allowed: 60

Score: /45

Percentage: /100

Question 1a

Propane and hexane are part of the alkane homologous series.

i)

Define the term *hydrocarbon*.

[1]

ii)

Give the general formula for the homologous series of alkanes.

[1]

iii)

State the formula of an **alkane** containing five carbon atoms.

[1]

[3 marks]

Question 1b

State **three** characteristics of a homologous series.

[3 marks]

Question 1c

Give the IUPAC names for the following aliphatic molecules in **Table 1.1**.

Table 1.1

Molecule	Name
$\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$	
$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$	
$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	
$\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_2\text{OH}$	

[4 marks]

Question 1d

The structures of four organic compounds are shown in Fig 1.1.

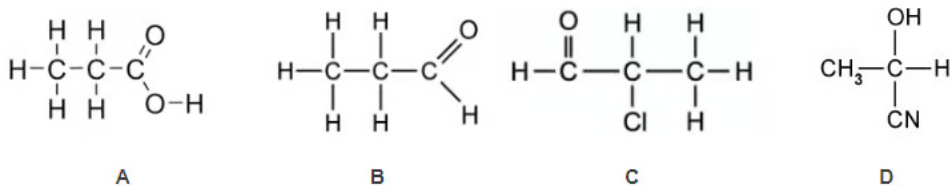


Fig 1.1

Complete Table 1.2 by naming the organic compounds and their functional groups.

Table 1.2

Organic compound	Name	Functional groups
A	Propanoic acid	
B		Aldehyde
C		Aldehyde AND Halogenoalkane
D	2-hydroxypropanenitrile	

[4 marks]

Question 1e

The compound 2,2-dimethylpentan-3-ol can be represented using different types of formulae.

Its structural formula can be shown as $\text{C}(\text{CH}_3)_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$.

i)

Give the empirical formula of 2,2-dimethylpentan-3-ol.

[1]

ii)

Draw the skeletal formula of 2,2-dimethylpentan-3-ol.

[1]

[2 marks]

Question 2a

State the **three** different types of structural isomerism.

[3 marks]

Question 2b

State the type of isomerism that occurs between the following pairs of compounds.

i)

2-methylpentane and 2,3-dimethylbutane.

[1]

ii)

Butan-1-ol and butan-2-ol

[1]

[2 marks]

Question 2c

Compounds **A** to **F** are shown in **Fig 2.1**.

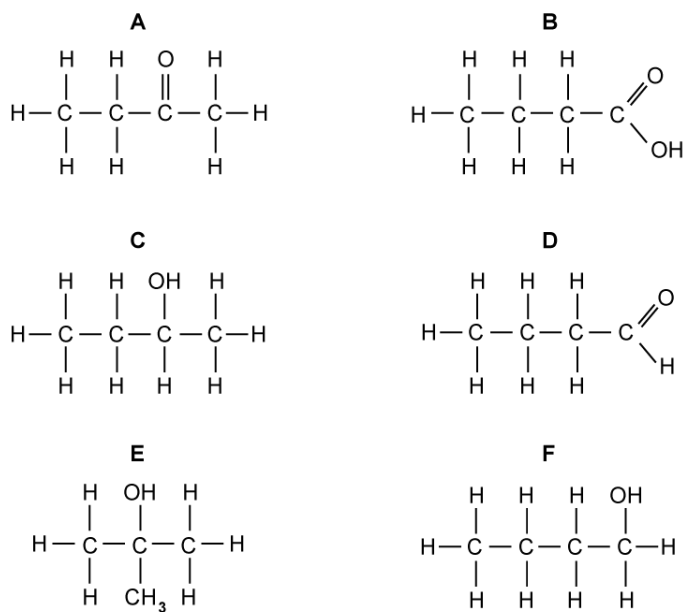


Fig 2.1

i)

State the IUPAC name for compound **B**.

[1]

ii)

State the type of structural isomerism that occurs between compounds **A** and **D**.

[1]

iii)

State the type of structural isomerism that occurs between compounds **C** and **E**.

[1]

[3 marks]

Question 3a

Free radical substitution reactions involve hydrogen atoms in alkanes being replaced by halogen atoms.

Name the **three** steps involved in a free radical substitution reaction.

[3 marks]

Question 3b

When a molecule of chlorine, Cl_2 , is exposed to UV light two chlorine radicals are formed.

i)

Write an equation for this reaction.

[1]

ii)

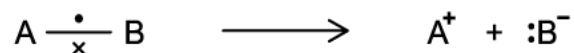
State the type of bond fission involved.

[1]

[2 marks]

Question 3c

Fig 1.1 shows the breaking of a covalent bond, where the more electronegative atom **B** has taken both electrons from the bond to form a negative ion.

Fig 1.1

State the name of this type of bond fission.

[1 mark]**Question 3d**

Name **three** other types of reaction mechanism.

[3 marks]**Question 4a**

Aliphatic compounds can be organised into three different categories depending on their structure.

One of the categories is straight-chain. Name the other **two** categories of aliphatic compounds.

[1 mark]

Question 4b

Complete Table 4.1 to show the type and number of bonds in hybridised bonds.

Table 4.1

Hybridisation	Number of σ bonds	Number of π bonds
sp^3		
sp^2		
sp		

[3 marks]

Question 4c

Four compounds are shown in Fig 4.1.

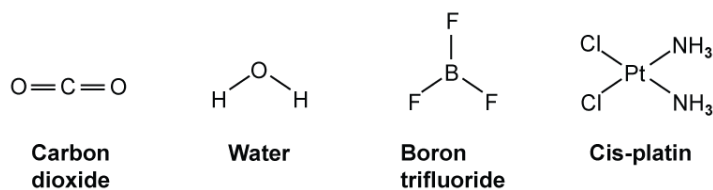


Fig 4.1

Complete Table 4.2 to name the shapes and bond angles of the **four** planar molecules.

Table 4.2

Molecule	Shape	Bond angle / °
Carbon dioxide	Linear	
Water		104.5
Boron trifluoride		120
Cis-platin	Square planar	

[4 marks]

Question 5a

Name the **two** types of stereoisomers.

[1 mark]

Question 5b

Describe what a chiral centre is.

[1 mark]

Question 5c

The structure of one optical isomer of a chlorofluorocarbon is shown in Fig 5.1.

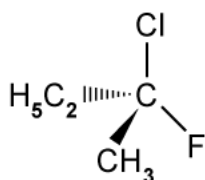


Fig 5.1

Draw the structure of the other enantiomer.

[1 mark]

Question 5d

The skeletal structure of a halogenoalkane is shown in Fig 5.2.

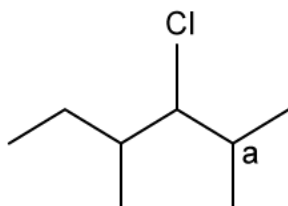


Fig 5.2

Explain why the carbon labelled **a** cannot be a chiral centre.

[1 mark]

Model Answers: Easy

1a

a)

i) The terms *hydrocarbon* and *homologous series* mean:

Hydrocarbon:

- A chemical that contains only hydrogen and carbon; [1 mark]

ii) The general formula for the homologous series of alkanes is:

- C_nH_{2n+2} ; [1 mark]

iii) The formula of an **alkane** with five carbon atoms is:

- C_5H_{12} ; [1 mark]

[Total: 3 marks]

- The definition of hydrocarbon can be a one or two mark question
 - You will not score maximum marks without the word 'only'
- You should know the general formula for the alkanes and alkenes specifically
- Part (iii) asks for the formula of the alkane, not the alkene
 - The general formula for an alkane is C_nH_{2n+2}
 - Therefore the formula for an alkane with five carbon atoms is C_5H_{12} as the number of hydrogen atoms is double the number of carbons and add 2
 - $(5 \times 2) + 2 = 12$

1b

b) Three characteristics of a homologous series are:

Any **three** of the following:

- Same chemical reactions / chemically similar; [1 mark]
- Same functional group; [1 mark]
- Same general formula; [1 mark]
- Increase by CH_2 / $M_r = 14$ (with each subsequent molecule); [1 mark]
- Have a trend in physical properties down the group e.g. boiling point; [1 mark]

[Total: 3 marks]

- The definition of homologous series is an important one to remember in order to get maximum marks
 - Don't forget a homologous series:
 - Contains the same functional group
 - Each subsequent compound in the group will differ by an additional CH_2 group
- There are lots of different homologous series you will study in this topic, so this definition is an important one that can apply to a range of organic compounds
- Don't be tempted just to say 'similar properties' for a characteristic of a homologous group:
 - This is too vague and you must either mention **similar chemical** or **similar physical properties** to get the mark
 - As you go down a homologous group, remember that the molecular and empirical formulae will change

1c

c) The IUPAC names for the following aliphatic molecules in **Table 1.1** are:

Molecule	Name
$\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$	2-chloropropane; [1 mark]
$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$	But-1-ene; [1 mark]
$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	Propan-1-ol; [1 mark]
$\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_2\text{OH}$	2-bromobutan-1-ol; [1 mark]

[Total: 4 marks]

- The basic principles for using IUPAC rules for naming these aliphatic molecules are:
 - Identify the longest unbranched chain of carbons, the stem is the name of the corresponding alkane
 - Identify the functional groups and alkyl side chains present, and select the appropriate prefixes and suffixes for them
 - Number the alkyl groups and functional groups to indicate their position on the longest unbranched chain
 - **Remember:** Number your groups to give you the lowest possible numbering

1d

d) The completed Table 1.2 with the names and functional groups of the four organic compounds is:

Organic compound	Name	Functional groups
A	Propanoic acid	Carboxylic acid; [1 mark]
B	Propanal; [1 mark]	Aldehyde
C	2-chloropropanal; [1 mark]	Aldehyde AND Halogenoalkane
D	2-hydroxypropanenitrile	Alcohol AND Nitrile; [1 mark]

[Total: 4 marks]

- This question is similar to part (c) but you are given displayed formulae instead of structural formulae and asked to name the functional groups
- Compound **A**
 - The functional group is the -COOH carboxylic acid
- Compound **B**
 - The functional group is the -CHO aldehyde group, which gives an -al ending to the name
 - The main structure contains three carbons (prop-) with only single bonds (-an-)
 - This means that the chemical name is propanal

- **Compound C**
 - The functional groups are
 - The -CHO aldehyde group, which gives the -al ending to the name **AND** automatically makes the -CHO carbon number 1
 - The -Cl group is on carbon number 2 in the chain, which contributes 2-chloro- to the name
 - The main structure contains three carbons (prop-) with only single bonds (-an-)
 - This means that the chemical name is 2-chloropropanal
- **Compound D**
 - The functional groups are:
 - The nitrile / CN group
 - The alcohol / OH group
 - The nitrile group is a higher priority in nomenclature which means that the alcohol group gives the hydroxy- prefix instead of the -ol suffix
 - **Careful:** If you were asked to name this compound, do not forget that there is a carbon in the nitrile group

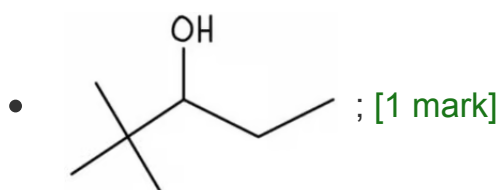
1e

e)

i) The empirical formula of 2,2-dimethylpentan-3-ol is:

- $\text{C}_7\text{H}_{16}\text{O}$; [1 mark]

ii) The skeletal formula of 2,2-dimethylpentan-3-ol is:



[Total: 2 marks]

- The molecular formula of 2,2-dimethylpentan-3-ol is $\text{C}_7\text{H}_{16}\text{O}$
 - **Remember:** The empirical formula of an organic compound gives the simplest whole number ratio of the different atoms of each component element
 - The molecular formula of 2,2-dimethylpentan-3-ol cannot be simplified

- This means that the empirical formula of 2,2-dimethylpentan-3-ol is the same as its molecular formula
- In a skeletal formula, the hydrogen atoms attached to carbons are generally omitted and the line ends / vertices represent carbon atoms
- For this question
 - The longest possible carbon chain is pentane
 - There are two methyl groups both attached to carbon-2
 - There is an alcohol group attached to carbon-3
 - **Careful:** The bond to the OH group must clearly be shown to the oxygen atom as this can sometimes cost you a mark in an exam

2a

a) The three different types of structural isomerism are:

- Chain (isomerism); [1 mark]
- Positional (isomerism); [1 mark]
- Functional group (isomerism); [1 mark]

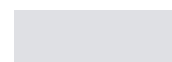
[Total: 3 marks]

- Structural isomers are compounds that have the same molecular formula but different structural formulae
- You should know the three different types and be prepared to determine the possible structural formulae of an organic molecule, given its molecular formula

2b

b)

i) The type of isomerism that occurs between 2-methylpentane and 2,3-



- There are two methyl groups both attached to carbon-2
- There is an alcohol group attached to carbon-3
- **Careful:** The bond to the OH group must clearly be shown to the oxygen atom as this can sometimes cost you a mark in an exam

2a

a) The three different types of structural isomerism are:

- Chain (isomerism); [1 mark]
- Positional (isomerism); [1 mark]
- Functional group (isomerism); [1 mark]

[Total: 3 marks]

- Structural isomers are compounds that have the same molecular formula but different structural formulae
- You should know the three different types and be prepared to determine the possible structural formulae of an organic molecule, given its molecular formula

2b

b)

i) The type of isomerism that occurs between 2-methylpentane and 2,3-dimethylbutane is:

- Chain (isomerism); [1 mark]

ii) The type of isomerism that occurs between butan-1-ol and butan-2-ol is:

- Positional (isomerism); [1 mark]

[Total: 2 marks]

- Chain isomerism is when compounds have the same molecular formula, but their

longest hydrocarbon chain is not the same length as there is branching present

- Positional isomers arise from differences in the position of a functional group in each isomer
 - The functional group can be located on different carbons, for example, butan-1-ol and butan-2-ol
 - Both compounds have an alcohol group and are made up of 4 carbons, 10 hydrogens and one oxygen, however, in butan-1-ol the functional group is located on the first carbon and in butan-2-ol on the second carbon

2c

c)

i) The name for compound **B** is:

- Butanoic acid; [1 mark]

ii) The type of structural isomerism shown by **A** and **D** is:

- Functional group isomerism; [1 mark]

iii) The type of structural isomerism shown by **C** and **E** is:

- Chain isomerism; [1 mark]

[Total: 3 marks]

- Compound **B** has four carbons in the chain (but-) with single carbon-carbon bonds (-an-) and a carboxylic acid functional group (-oic acid)
 - These pieces combine to give the chemical name butanoic acid
- **A** and **D** are functional group isomers
 - **A** is a ketone, (butanone) and **D** is an aldehyde (butanal)
 - When the location of the C=O bond changes from the middle of the chain to the end of the chain, the compound becomes an aldehyde
 - The number of carbon atoms and hydrogen atoms has not changed
 - They are both C₄H₈O
- **C** and **E** are chain isomers
 - They are both C₄H₁₀O
 - The alcohol group remains on the second carbon in the chain
 - There is no chain in the functional groups

- The carbon chain has shortened as a -CH_3 and a hydrogen atom have swapped places

3a

a) The **three** steps involved in a free radical substitution reaction are:

- Initiation; [1 mark]
- Propagation; [1 mark]
- Termination; [1 mark]

[Total: 3 marks]

- You need to know the steps involved in a free radical substitution reaction
- Lots of students find the propagation steps the most challenging

3b

b)

i) The equation is:

- $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$; [1 mark]

ii) The type of bond fission is:

- Homolytic (bond fission); [1 mark]

[Total: 2 marks]

- When a molecule of Cl_2 is exposed to UV light it will undergo homolytic fission
 - $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
- For homolytic fission, a single-headed arrow is required
- This produces two radicals, $\text{Cl}\cdot$
- Each atom gets one of the two electrons
- A (free) radical is a species with one (or more) unpaired electrons

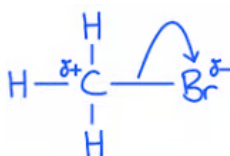
3c

c) This type of fission is:

- Heterolytic (fission); [1 mark]

[Total: 1 mark]

- In this type of bond fission, one atom takes both original bonding electrons from the other atom
- An example of heterolytic fission is the breaking of the C-Br bond in a halogenoalkane
 - $\text{CH}_3\text{Br} \rightarrow \text{CH}_3^+ + \text{Br}^-$
- Under certain conditions, the electrons will move from the C-Br bond to the more electronegative Br atom
- We show this by using a double-headed curly arrow as shown in the diagram below:



- This produces CH_3^+ and Br^-
- **Remember:**
 - A double-headed, curly arrow shows the movement of a pair of electrons
 - A single-headed curly arrow shows the movement of a single electron

3d

d) **Three** other types of reaction mechanism are:

Any **three** of the following:

- Electrophilic addition; [1 mark]
- Nucleophilic substitution; [1 mark]
- Elimination; [1 mark]
- Nucleophilic addition; [1 mark]
- Electrophilic substitution; [1 mark]
- Nucleophilic addition-elimination; [1 mark]

[Total: 3 marks]

- Free radical substitution is not an accepted answer as it was given to you in part (a)
- You should know the mechanisms all of which involve the movement of electrons

and can be represented by curly arrows

- Electrophilic addition, nucleophilic substitution and elimination are the curly arrow mechanisms that are required for the AS course
 - Electrophilic addition reactions - where two reactants combine to form one product, e.g. ethene and bromine undergoing addition to form 1,2-dibromoethane
 - Nucleophilic substitution reactions - where an atom or group of atoms in a compound is replaced by another atom or group of atoms, e.g. bromoethane reacting with the hydroxide ion to form ethanol and the bromide ion
 - Elimination reactions - where a small molecule is removed from a larger molecule, e.g. ethanol reacting with an acid catalyst to form ethene alongside a small molecule of water which is eliminated

4a

a) The **two** other categories of aliphatic compounds are:

- Branched
AND
Cyclic; [1 mark]

[Total: 1 mark]

- You should know these three types of compound as they can help you with isomers

4b

b) The completed Table 4.1 showing the type and number of bonds in hybridised bonds is:

Hybridisation	Number of σ bonds	Number of π bonds
sp^3	1	0

AND

Cyclic; [1 mark]

[Total: 1 mark]

- You should know these three types of compound as they can help you with isomers

4b

b) The completed Table 4.1 showing the type and number of bonds in hybridised bonds is:

Hybridisation	Number of σ bonds	Number of π bonds
sp^3	1	0
sp^2	1	1
sp	1	2

- $sp^3 = 1$ sigma / σ bond

AND

0 pi / π bonds; [1 mark]

- $sp^2 = 1$ sigma / σ bond

AND

1 pi / π bond; [1 mark]

- $sp = 1$ sigma / σ bond

AND

2 pi / π bonds; [1 mark]

[Total: 3 marks]

- sp^3 is essentially a single bond
 - This means that it is only one sigma / σ bond
- sp^2 is a double bond
 - This means that there is one sigma / σ bond
 - The second part of the double bond is one pi / π bond
- sp is a triple bond
 - This means that there is one sigma / σ bond
 - The remainder of the triple bond is two pi / π bonds

4c

c) The completed Table 4.2 to name the shapes and bond angles of the **four** planar molecules is:

Molecule	Shape	Bond angle / °
Carbon dioxide	Linear	180; [1 mark]
Water	Non-linear OR Bent; [1 mark]	104.5
Boron trifluoride	Trigonal planar; [1 mark]	120
Cis-platin	Square planar	90; [1 mark]

[Total: 4 marks]

- Make sure that you know your shapes and bond angles of molecules
- Examiners often comment on how poorly questions about bond angles, in particular, are answered

5a

a) The **two** types of stereoisomers are:

- Geometric
AND
Optical; [1 mark]

[Total: 1 mark]

- Stereoisomers are molecules with the same structural formula but a different arrangement of atoms in 3D space

5b

b) A chiral centre is:

- A carbon atom that has four different atoms / groups of atoms attached to it; [1 mark]

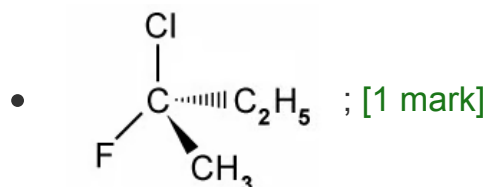
[Total: 1 mark]

- Defining a chiral centre is not a common question because the focus is on identifying and drawing them

- But, you cannot identify or draw them without knowing this definition

5c

c) The structure of the other enantiomer is:



[Total: 1 mark]

- **Remember:** Enantiomers / optical isomers are non-superimposable mirror images
 - This means that you just have to draw the mirror image of the structure in the question

5d

d) The carbon labelled **a** cannot be a chiral centre because

- It has two methyl groups

OR

It does not have four different groups attached to it; [1 mark]

[Total: 1 mark]

- **Remember:** Chiral centres have four different atoms / groups of atoms attached

